THE DEDEKIND-FINITE TREE U_T FALLS INTO THE DEFINABLE CARDINALITY CLASS DELTA_5

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Allowed standard prerequisites (editorial).

- The FM-model with finite supports built from the automorphism group of the tree, and the definable closure operation on it. Set up in the paragraphs before the counted statement (source lines 303-305) and used throughout the counted proof. Supplies the model in which the cardinality comparison

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is made and the closure operator with which the support of a function is measured.
- The tree-theoretic description of [A union {x}] as [A] together with the new point and the meets it creates. Derived inside the counted proof at lines 317-319. Is what lets the two case analyses conclude that a point outside [A] is fixed by the function.

Theorem 0.1. *For any countable weakly 2-transitive tree T, $|U_T| \in \Delta_5$ in $\mathfrak{N}_T$, and if T has infinitely many ramification orders, then $|U_T| \notin \Delta_4$.*

Proof. Let $f : U_T \to U_T \cup \{*\}$ where $* \notin U_T$ be surjective in $\mathfrak{N}_T$ with finite support A.

We remark that for any x, y, if $f(x) = y$, then $y \in [A \cup \{x\}]$. For since A supports f, $\pi(f) = f$ for all $\pi \in G_A$. Then $\langle x, y\rangle \in f$ and so $\langle \pi x, \pi y\rangle \in \pi(f) = f$. Hence if $\pi y \neq y$, then $\pi x \neq x$ since f is a function. If $y \notin [A \cup \{x\}]$, then by definition of definable closure [ ], there is $\pi \in G_{A\cup\{x\}} \subseteq G_A$ such that $\pi y \neq y$ but $\pi x = x$, contrary to what we have shown. Hence $y \in [A \cup \{x\}]$. Also note that $[A \cup \{x\}]$ is the union of $[A] \cup \{x\}$ and $\{x \wedge a : a \in A\}$, together with the minimal points of cones at exceptional ramification points in this set.

Now consider $y \notin [A]$ and let x be such that $f(x) = y$. We aim to show that $x = y$. Now $x \in [A]$ would imply that $y \in [A]$, contrary to supposition, and we deduce that $x \notin [A]$.

Case 1: $y \not\leq a$ for all $a \in [A]$.

Then $y \neq x \wedge a$ for all $a \in A$, and also y is not the minimal point of a cone at an exceptional ramification point of $[A]$ (as this would lie in $[A]$). So if $y \neq x$, then y must be the minimal point of a cone at an exceptional ramification point of the form $x \wedge a$ for some $a \in A$.

Let z be such that $f(z) = x$. If $x \leq b$ for some $b \in [A]$, then $x \wedge a = b \wedge a$, which is impossible. Applying the same argument as above for x and z, x must be the minimal point of a cone at an exceptional ramification point of the form $z \wedge b$ for some $b \in A$. But now again $x \wedge a = b \wedge a$ which is impossible.

The conclusion is that $x = y$.

Case 2: $y \leq a$ for some $a \in [A]$.

If $x \not\leq b$ for all $b \in [A]$, then by the argument in Case 1, $f(x) = x$, and $x = y$ again follows. Hence $x \leq b$ for some $b \in A$. Then for any $c \in A$, either $x \wedge c = x$ or $x \wedge c \in [A]$. Since $x \notin [A]$, no new ramification points are added in passing from $[A]$ to $[A \cup \{x\}]$, and therefore $[A \cup \{x\}] = [A] \cup \{x\}$. It follows that $x = y$.

From both cases, we have $f(y) = y$ for all $y \notin [A]$. Hence $f[A] = [A] \cup \{*\}$ which is a contradiction since f is surjective but $[A]$ is finite. Therefore such f does not exist in $\mathfrak{N}_T$ and so $|U_T| \in \Delta_5$ in $\mathfrak{N}_T$.

The final clause follows from the fact that T is not $\aleph_0$-categorical. □